



Bridging the Lab-to-Field gap in plant disease diagnosis through unsupervised domain adaptation enhanced by background recomposition

Woosang Jeon^{a,b}, Taehyeong Kim^{a,b,c,*}, Sanghyeok Choi^{a,b}, Kyuseok Yang^{a,b},
Seong-Yeop Kim^{b,d}, Mungyeong Song^{b,d}

^a Department of Biosystems Engineering, Seoul National University, Seoul 08826, Republic of Korea

^b Integrated Major in Global Smart Farm, Seoul National University, Seoul 08826, Republic of Korea

^c Research Institute of Agriculture and Life Sciences, Seoul National University, Seoul 08826, Republic of Korea

^d Graduate School of International Agricultural Technology, Seoul National University, Gangwon-do 25354, Republic of Korea

ARTICLE INFO

Dataset link: <https://doi.org/10.5281/zenodo.17901489>, <https://github.com/LABA-SNU/FBR-UDA>

Keywords:

Plant disease diagnosis
Domain discrepancy
Labeled data scarcity
Unsupervised domain adaptation
Background recomposition

ABSTRACT

Early detection of plant diseases caused by pests, pathogens such as viruses is essential for preventing their spread and minimizing crop damage, particularly in indoor farming and orchards. Machine learning-based computer vision approaches have emerged as promising tools for automated disease diagnosis, but their implementation faces critical challenges due to the scarcity of labeled data from real agricultural environments. Consequently, models are primarily trained using laboratory-collected images with monotonous backgrounds, resulting in significantly reduced performance when deployed in real cultivation environments. To address these challenges, this study presents a two-step adaptation method to bridge the gap between laboratory and real-field environments in plant disease diagnosis. Our approach first applies field-adaptive background recomposition for image augmentation, followed by unsupervised domain adaptation, enabling effective disease diagnosis in real agricultural environments. We quantitatively and qualitatively validated the proposed method on disease identification tasks for tomato, chili pepper, grape, and apple leaves, achieving robust performance in field environments without requiring labeled real-field data.

1. Introduction

In cultivation environments such as smart farms and orchards, early diagnosis and control of plant diseases are crucial to maintaining crop health and productivity (Martinelli et al., 2015; Yashodha and Shalini, 2021; Elumalai et al., 2024). These measures are essential to minimize crop damage by controlling the rapid spread of pathogens, which can otherwise lead to significant losses. One promising approach involves the use of computer vision models, which excel in extracting features from images for swift and accurate disease diagnosis.

For example, numerous studies have demonstrated the potential of computer vision models in the diagnosis of plant disease. These models have shown remarkable performance in classifying multiple diseases across various crops (Mohanty et al., 2016; Liu and Wang, 2021). Some researchers have achieved high accuracy rates in identifying dozens of diseases in different plant species, while others have focused on improving the efficiency and speed of diagnosis (Grinblat et al., 2016; Fan et al., 2022). In addition, efforts have been made to develop models capable of distinguishing between subtle variations of similar diseases (Yan et al., 2021; Wu et al., 2023), assessing disease severity (Yu

et al., 2025), and integrating image-text modalities using textual trait descriptions to enhance disease diagnostics (Zhao et al., 2024). These advances in computer vision applications for plant pathology have consistently yielded promising results, particularly in controlled settings where many models achieve accuracy rates exceeding 90% (Liu and Wang, 2020; Kaushik et al., 2020; Shoaib et al., 2022; Syed-Ab-Rahman et al., 2022).

However, deploying these computer vision technologies in the agricultural domain faces two significant constraints. First, there is a notable scarcity of images depicting infected specimens, which are essential for effectively training machine learning models (Ramesh et al., 2018). This shortage impedes the development of robust and reliable disease classification systems. Second, accurately labeling the presence of disease in these images is challenging without the expertise of plant and crop specialists (Mahlein, 2016). This lack of specialized knowledge hinders the development of robust diagnostic models adaptable to diverse cultivation environments, as accurate identification relies on expert interpretation of subtle disease indicators.

* Corresponding author at: Department of Biosystems Engineering, Seoul National University, Seoul 08826, Republic of Korea.
E-mail address: taehyeong.kim@snu.ac.kr (T. Kim).

Nomenclature	
Abbreviations	
AI-Hub-L	Laboratory-like subset of AI-Hub (background-removed images)
AI-Hub-R	Real-field subset of AI-Hub (images captured in actual cultivation environments)
CDAN+E	Conditional Domain Adversarial Network with Entropy Conditioning
CORAL	CORrelation ALignment
DALN	Discriminator-free Adversarial Learning Network
DANN	Domain-Adversarial Neural Network
DDC	Deep Domain Confusion
DCORAL	Deep CORrelation ALignment
FBR	Field-adaptive Background Recomposition
GRL	Gradient Reversal Layer
JSD	Jensen–Shannon Divergence
NWD	Nuclear Wasserstein Discrepancy
MMD	Maximum Mean Discrepancy
PPD	Plant-Pathology Dataset (real-field images)
PVD	PlantVillage Dataset (laboratory images)
SAM	Segment Anything Model
t-SNE	t-distributed Stochastic Neighbor Embedding
UDA	Unsupervised Domain Adaptation
Symbols	
D	Domain discriminator in adversarial domain adaptation
d_f	Dimension of the feature space
G	Label classifier mapping features to predicted labels
I	Number of iterations for morphological operations
k	Kernel size for morphological operations (erosion, dilation and Gaussian blur)
$\ \cdot \ _F$	Frobenius norm measuring element-wise differences between matrices
$\ \cdot \ _*$	Nuclear norm (sum of singular values of a matrix)
$\mathcal{L}_{\text{CORAL}}$	CORAL loss function for aligning second-order feature statistics
$\mathcal{L}_{\text{NWD}}$	NWD loss function for aligning feature distributions
$\mathcal{L}_{\text{MMD}}$	MMD loss function for aligning feature distributions
N_s	Number of samples in the source domain
N_t	Number of samples in the target domain
P_s	Softmax probability matrix for the source
P_t	Softmax probability matrix for the target
$\Phi(\cdot)$	Feature extraction function mapping input images to feature vectors
Σ	Summation operator; e.g., Σ_{ϕ_s} , Σ_{ϕ_T}
S_i	The i th sample from the source domain (e.g., image)
T_j	The j th sample from the target domain (e.g., image)
X_S	Input samples from the source domain (drawn as mini-batches during training)
X_T	Input samples from the target domain (drawn as mini-batches during training)

Consequently, the development and training of plant disease classification models mainly rely on datasets such as PlantVillage dataset (PVD) (Hughes et al., 2015), which consist of images from laboratory settings. Additionally, for certain diseases with subtle symptoms,

particularly viral infections, researchers often generate training data through controlled inoculation of plants in laboratory environments to ensure accurate ground truth labeling while maintaining biosafety protocols (Zhu et al., 2016, 2017; Nguyen et al., 2021). These controlled images exhibit uniform backgrounds and are captured from consistent angles, starkly contrasting with the diverse and variable conditions encountered in actual agricultural environments. The Lab-to-Field gap encompasses multiple dimensions beyond background variations, including dynamic lighting changes (e.g., shadows, overexposure, and seasonal illumination variations) and variations in leaf orientations and imaging conditions. These multifaceted environmental factors collectively contribute to significant domain discrepancies that challenge model generalization.

Fig. 1 illustrates the significant discrepancies between laboratory and real-field conditions in the context of plant disease diagnosis models, demonstrating how these multidimensional domain gaps manifest in practical applications. This comparison highlights the domain differences and consequent performance degradation when laboratory-trained models are applied to real-world agricultural settings, exposing the challenges in generalizing these models to diverse field conditions. These findings underscore the importance of developing adaptive models capable of maintaining diagnostic reliability and effectiveness across various agricultural environments that differ substantially from controlled laboratory conditions.

Unsupervised domain adaptation (UDA) methods have been applied across various agricultural tasks to enhance the adaptability of AI models in diverse agricultural environments (Kim et al., 2023; Zhao et al., 2023). Marino et al. (2020) implemented UDA for potato defect detection, simulating domain changes through different potato varieties and image conditions. Regarding plant counting, Shi et al. (2022) utilized UDA to differentiate background from weeds. Similarly, Ilyas et al. (2023) applied UDA to improve model robustness in crop-weed recognition tasks. In the context of plant disease diagnosis, Wu et al. (2023) found that even with substantial real-world data for UDA, experiments on the task of apple leaf disease classification achieved only around 72% accuracy, indicating the need for more robust adaptation methods in complex agricultural settings.

Addressing these limitations, we propose a method that effectively handles limited labeled data from natural agricultural environments. Our approach first incorporates an image augmentation technique that integrates field-like backgrounds into laboratory images. For this process, we separate leaf regions using the Segment Anything Model (SAM) (Kirillov et al., 2023), an off-the-shelf instance segmentation model. To further mitigate remaining discrepancies between augmented and real-field images, it then employs distribution-based and adversarial UDA methods. This two-step process aims to reduce the feature disparity between laboratory images and those from actual agricultural settings.

To validate the effectiveness of our proposed method across diverse crops and environmental conditions, we first conducted experiments in an emulated setting by removing backgrounds from greenhouse images to create laboratory-like conditions. This systematic approach allows us to evaluate the domain adaptation process with known ground truth, as the corresponding plant diseases and crop varieties are maintained in both background-removed laboratory-like images (i.e., source domain) and their original real-field versions (i.e., target domain). Subsequently, we validated our proposed method on datasets representing genuine settings, where the domain gap stems from inherent differences in leaf characteristics, lighting conditions, and image acquisition processes between laboratory and field environments. Through quantitative and qualitative evaluations on both emulated and genuine settings, our method effectively mitigated the performance degradation typically observed when models trained on laboratory data are applied to real-field scenarios.

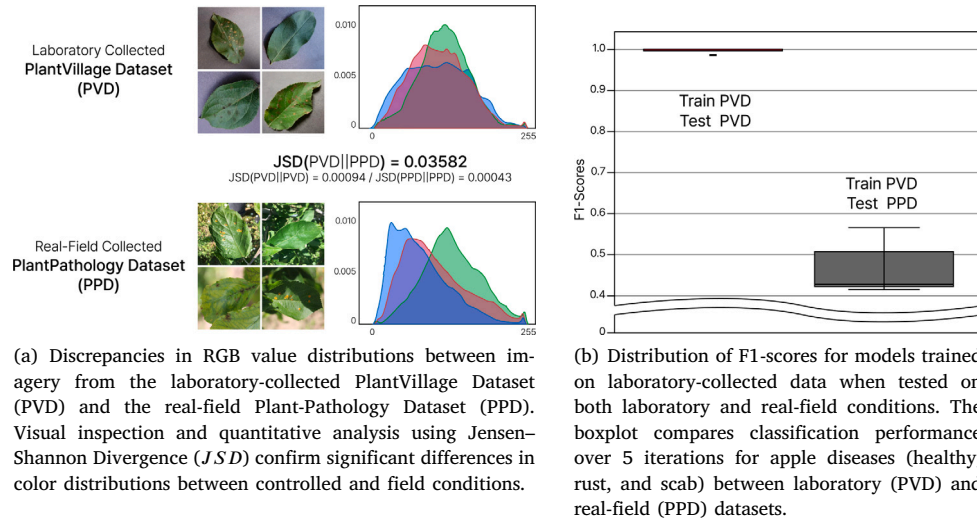


Fig. 1. Performance degradation in models trained on lab-collected data applied to real-field conditions. The discrepancies between laboratory and real-field data result in severe performance degradation when models trained on laboratory-collected data are applied to real-field conditions, highlighting a critical challenge in practical applications of plant disease classification.

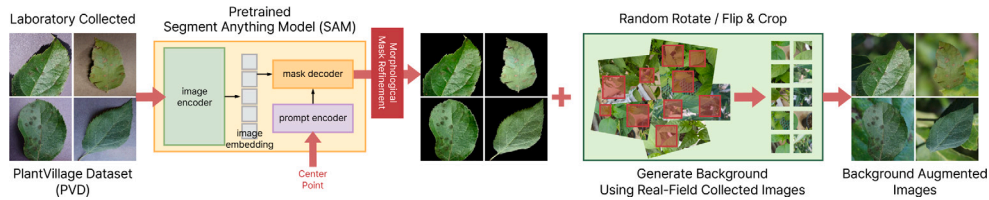


Fig. 2. The process of field-adaptive background recomposition (FBR) using real-environment images. PVD images, collected in a uniform laboratory setting, undergo leaf segmentation using SAM with central point prompts. The resulting masks are refined using traditional image processing techniques. Concurrently, backgrounds from real-field images are randomly rotated and cropped. Finally, isolated leaf regions are composited onto these backgrounds, creating augmented images that resemble real-field conditions.

2. Methodology

2.1. Field-adaptive background recomposition

Laboratory-collected images (e.g., PVD) are typically acquired under consistent and simplified background and lighting conditions, whereas real-field images exhibit highly diverse backgrounds and varying illumination. This discrepancy in image characteristics is a significant factor contributing to performance degradation when models trained on laboratory data are applied to real-world scenarios. To address this issue, we extract leaf regions from the laboratory-collected images and recompose them with real-field based backgrounds as illustrated in Fig. 2.

This process of field-adaptive background recomposition (hereinafter FBR) begins with automated and efficient leaf segmentation, performed using the pretrained SAM. Given that the leaf subjects in the laboratory-collected images are consistently positioned at the center, we utilize the image's central point coordinates as input prompts for SAM, allowing it to generate reliable segmentation masks with minimal manual intervention.

After obtaining the initial segmentation masks from SAM, we apply a morphological post-processing pipeline to refine the masks, addressing both segmentation artifacts and boundary preservation (Fig. 3). Initially, a dilation operation with a kernel size of k_d is performed to unify fragmented leaf regions and eliminate segmentation artifacts. This is followed by an erosion operation with a kernel size k_e , which is slightly smaller than that used for dilation (k_d), to suppress undesirable noise and adjust the mask so that the entire leaf region is fully covered with intact edges. These operations are iteratively applied over I

iterations, progressively refining the mask by filling incomplete regions and suppressing residual boundary noise while preserving the integrity of leaf contours. Following this refinement, a Gaussian blur with kernel size (k_g) is applied to smooth any remaining discontinuities, facilitating a seamless transition between leaf boundaries and the background. Although SAM provides a strong zero-shot segmentation capability, its outputs may exhibit minor imperfections along object boundaries. By incorporating post-processing techniques, our approach enhances segmentation accuracy and boundary definition without requiring manual corrections. As a result, the final masks exhibit improved consistency and realism, improving the quality of background recomposition for domain adaptation.

Using the refined segmentation masks, we extract the leaf regions from laboratory images. These extracted leaf images are then combined with diverse background images which are generated by randomly rotating and cropping unlabeled images collected from real-fields. This integration process results in background-augmented images that not only enrich the training dataset but also are visually closer to actual field conditions, thereby enhancing the model's robustness and generalization capabilities.

2.2. Unsupervised domain adaptation

Despite the application of background augmentation techniques, notable discrepancies persist between the augmented laboratory images and real-world images. These disparities manifest in several forms, encompassing artifacts at leaf edge boundaries, variations in color tones and shading, differences in texture details, and disparities in lighting conditions. To address these residual domain gaps, we implement

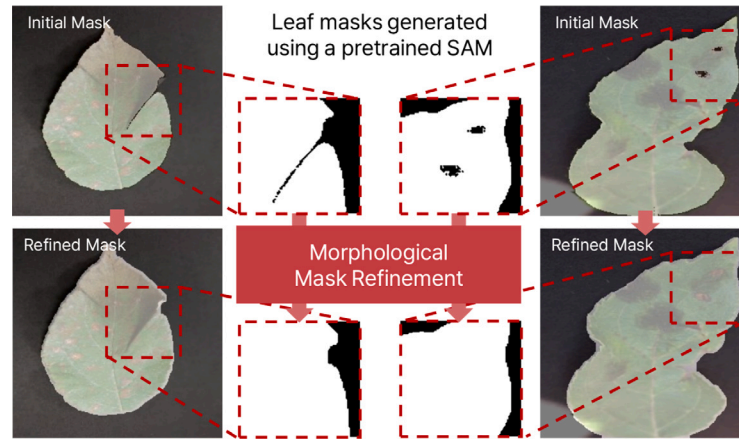


Fig. 3. Examples of morphological mask refinement. The top row shows the initial masks generated by a pretrained SAM, which contain small holes and segmentation artifacts. The bottom row presents the refined masks after applying morphological operations, including dilation to fill holes, erosion to suppress noise, and Gaussian blurring to smooth edges. These refinements result in more coherent segmentation with well-defined boundaries and fewer artifacts, leading to improved accuracy in leaf segmentation.

a range of unsupervised domain adaptation frameworks, incorporating both distribution-based and adversarial methodologies. Specifically, we employ Deep Domain Confusion (DDC) (Tzeng et al., 2014) and Deep Correlation Alignment (DCORAL) (Sun and Saenko, 2016) as distribution-based approaches, while utilizing Domain-Adversarial Neural Networks (DANN) (Ganin et al., 2016) and Conditional Domain Adversarial Networks with Entropy Conditioning (CDAN+E) (Long et al., 2018) as adversarial methods. We also consider the Discriminator-free Adversarial Learning Network (DALN) (Chen et al., 2022), which represents a hybrid approach between distribution-based and adversarial approaches.

2.2.1. Distribution-based domain adaptation

Distribution-based domain adaptation methods (i.e., DDC and DCORAL) focus on aligning feature distributions between source and target domains, which is crucial for adapting models trained on laboratory data to real-field agricultural settings. DDC introduces a domain confusion loss based on the Maximum Mean Discrepancy (MMD). MMD quantifies the difference between two probability distributions by comparing the means of their implicit feature mappings in a high-dimensional space. In this context, it measures the discrepancy between the feature distributions of the source and target domains. The MMD loss is defined as:

$$\mathcal{L}_{\text{MMD}} = \left\| \frac{1}{N_s} \sum_{i=1}^{N_s} \Phi(\mathbf{S}_i) - \frac{1}{N_t} \sum_{j=1}^{N_t} \Phi(\mathbf{T}_j) \right\|^2,$$

where $\Phi(\cdot)$ represents the feature extraction function, mapping input data to a learned feature space. $\mathbf{S}_i$ and $\mathbf{T}_j$ denote individual input samples from the source and target domains, respectively. N_s and N_t represent the number of samples in the source and target domains. By minimizing this MMD loss, DDC aims to reduce the disparity between the feature distributions of the source and target domains. This process encourages the learned features to be domain-invariant, thereby facilitating effective domain adaptation.

DCORAL extends the Correlation Alignment (CORAL) technique (Sun et al., 2016) to deep neural networks, aiming to align the second-order statistics of feature distributions between source and target domains. These statistics, represented by covariance matrices, capture the relationships and correlations among different features. The CORAL loss is then defined as:

$$\mathcal{L}_{\text{CORAL}} = \frac{1}{4d_f^2} \left\| \Sigma_{\Phi_S} - \Sigma_{\Phi_T} \right\|_F^2,$$

where Σ_{Φ_S} and Σ_{Φ_T} denote the covariance matrices of the feature distributions extracted by Φ from the source and target domains,

respectively, and d_f represents the dimension of the feature space. $\|\cdot\|_F$ denotes the Frobenius norm, which measures the element-wise difference between matrices. By minimizing this loss, DCORAL aims to align the second-order statistics of the extracted feature distributions. This alignment reduces the difference in feature relationships between domains, potentially improving the model's ability to generalize across different domains. As a result, the learned representations become less sensitive to domain-specific variations, enhancing the model's performance on the target domain.

2.2.2. Adversarial domain adaptation

In addition to distribution-based approaches, this study explores adversarial domain adaptation methods (i.e., DANN and CDAN+E). DANN architecture integrates three key components: a feature extractor Φ , a label classifier G , and a domain discriminator D , as illustrated in Fig. 4(a). These structures aim to learn domain-invariant features through adversarial training. To this end, the feature extractor learns to extract salient characteristics for disease recognition, while the domain discriminator differentiates between lab-generated and real-field images. The core mechanism of adversarial training is implemented through a gradient reversal layer (GRL) integrated between the feature extractor and the domain discriminator. This specialized layer reverses gradient flow during backpropagation, encouraging the feature extractor to minimize domain discrepancies by hindering the discriminator's ability, while preserving its capacity to extract disease-relevant features.

CDAN (Long et al., 2018) builds upon the DANN architecture, enhancing it by conditioning the domain discriminator on the cross-product of the feature representations and the predictions of the label classifier. This multilinear conditioning enables the model to capture more nuanced relationships between extracted features and their corresponding labels, potentially facilitating more effective domain adaptation. CDAN+E, illustrated in Fig. 4(b), is a variant of CDAN that incorporates entropy conditioning, a mechanism designed to guide the domain discriminator to emphasize samples with more confident label predictions. This approach employs an entropy-based weighting scheme, assigning greater importance to samples with lower prediction entropy, i.e., those with more certain label classifications. Through this mechanism, CDAN+E aims to enhance the domain discriminator's ability to identify genuine domain-specific differences, potentially mitigating the impact of ambiguous or outlier samples.

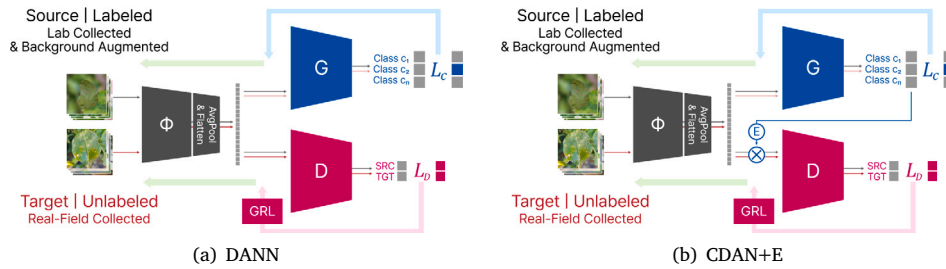


Fig. 4. The architectures of unsupervised domain adaptation methods based on adversarial networks. Both DANN (a) and CDAN+E (b) architectures consist of a feature extractor (Φ), a label classifier (G), and a domain discriminator (D). The GRL reverses the gradient flow from the domain discriminator, helping to remove domain-specific features while retaining class-discriminative features, thus enhancing the model's ability to generalize across domains. CDAN+E extends DANN by conditioning the domain discriminator on the cross-product of features and classifier predictions, with an additional entropy-based weighting mechanism to focus on more confident examples.

2.2.3. Hybrid distribution–adversarial adaptation

We also consider DALN (Chen et al., 2022), which reuses the task-specific classifier outputs to measure domain discrepancy. DALN introduces the Nuclear Wasserstein Discrepancy (NWD) to quantify domain shift, computed from the nuclear norms of the class probability matrices for source and target domains:

$$\mathcal{L}_{\text{NWD}} = \frac{1}{N_t} (-\|P_T\|_* + \|P_S\|_*),$$

where

$$P_S = \text{softmax}(f_\theta(\text{GRL}(\Phi(X_S)))), \quad P_T = \text{softmax}(f_\theta(\text{GRL}(\Phi(X_T)))).$$

Here, $\Phi(\cdot)$ denotes the feature extractor, f_θ is the task-specific classifier parameterized by θ , and GRL represents the gradient reversal layer, which acts as an identity mapping during the forward pass but reverses the sign of gradients in backpropagation. Each row of P_S or P_T corresponds to the softmax output of an individual sample, with X_S and X_T denoting mini-batches of source and target samples, respectively. N_t is the number of target samples in the batch, and $\|\cdot\|_*$ denotes the nuclear norm (the sum of singular values).

This formulation penalizes the discrepancy in predicted class distributions between domains, thereby encouraging alignment in the classifier's output space. Through the adversarial sign-reversal mechanism of GRL, the NWD loss is propagated into the feature extractor while remaining directly grounded in distributional alignment. In this sense, DALN occupies an intermediate position between distribution-based and adversarial approaches: it inherits the explicit distribution-matching property of the former while leveraging the adversarial training signal of the latter to promote domain-invariant feature learning. In practice, DALN is jointly optimized with the standard supervised classification loss on the source domain, ensuring discriminative learning from labeled source samples while simultaneously aligning source and target distributions through NWD.

3. Experiments

3.1. Datasets

Our study utilizes three distinct datasets for plant diseases diagnosis: the AI-Hub¹ dataset collected from real cultivation environments for emulated experiments, and two more datasets for the genuine application — the laboratory-collected PlantVillage² dataset (PVD) and the real-field-collected PlantPathology³ dataset (PPD).

The AI-Hub dataset offers a comprehensive collection of agricultural images across 12 different crops, all captured in real greenhouse

environments. For our study, we examined the leaf image subsets of tomato, chili pepper, and grape. To create our emulated setting, we processed the AI-Hub images by removing their backgrounds, creating AI-Hub-L (representing laboratory-like conditions) as our source domain. The original, unprocessed images, denoted as AI-Hub-R, were used as the target domain, preserving their real-world environmental characteristics.

In contrast, the genuine setting utilized two different data sources: the laboratory-collected PVD dataset as the source domain and the real-field PPD dataset as the target domain. The PVD dataset, a widely recognized benchmark in plant disease diagnosis research, comprises over 54,000 images across 38 classes of various crops and diseases, captured in controlled laboratory settings with plain backgrounds. From this dataset, we extracted the subset of apple leaf images with disease labels corresponding to those in the PPD dataset. The PPD dataset consists of approximately 4900 apple leaf images captured in real orchard environments. This dataset represents authentic field conditions, including varying light conditions, leaf orientations, and physiological ages of leaves, making it particularly valuable for evaluating domain adaptation methods under real-world environments. The selected crop datasets contain the following disease categories:

- **Tomato:** healthy, leaf mold, and Yellow Leaf Curl Virus
- **Chili pepper:** healthy, leaf spot, and Pepper Mild Mottle Virus
- **Grape:** healthy and downy mildew
- **Apple:** healthy, cedar-apple rust, and leaf scab

For model training, we used labeled data from AI-Hub-L and PVD as source domains in the emulated and genuine settings, respectively. From the AI-Hub-R and PPD datasets, we designated separate sets of images for target domains, which remained unlabeled during training to maintain the unsupervised nature of domain adaptation. We also prepared independent test sets using distinct images with labels for quantitative evaluation. The remaining real-field images from these datasets were used for the proposed background augmentation method. Table 1 presents the specific number of images used for source domains, target domains, and test sets in our study, where each dataset was carefully partitioned to ensure no overlap between training and evaluation data.

3.2. Implementation details

We applied background augmentation to the entire source domain dataset, where the initial SAM-generated masks were used to separate leaf regions from their original backgrounds. In this step, we performed morphological operations to improve segmentation quality by refining these masks and enhancing boundary smoothness. Specifically, we applied two iterations ($I = 2$) of morphological refinement using dilation ($k_d = 15$) and erosion ($k_e = 10$), followed by Gaussian blurring ($k_g = 5$), which were empirically determined through visual assessment. The

¹ <https://aihub.or.kr>.

² <https://github.com/spMohanty/PlantVillage-Dataset>.

³ <https://www.kaggle.com/c/plant-pathology-2020-fgvc7>.

Table 1

Composition and number of samples in experimental datasets. For the emulated setting, AI-Hub-L serves as the source domain while AI-Hub-R provides target and test data. For the genuine setting, labeled PVD data serves as the source domain, with PPD data serving as target and test domains. Note that all target domain datasets remain unlabeled during the domain adaptation process.

	Emulated				Genuine	
	Dataset	Tomato	Chili Pepper	Grape	Dataset	Apple
Source (Laboratory)	AI-Hub-L	1320	1000	216	PVD	825
Target (Real-Field)	AI-Hub-R	990	750	162	PPD	900
Test (Real-Field)	AI-Hub-R	363	512	62	PPD	600

extracted leaf regions were then recomposited onto background images sampled from the target domain.

From this augmented dataset, we reserved 25% for validation, using it for model selection, while the remaining 75% served as our training data. This approach ensured that both our training and validation data underwent the same augmentation process, allowing us to assess model performance consistently while maintaining separation from the test set of target domain.

For the model architecture, we utilized ResNet-18 (He et al., 2016), pre-trained on ImageNet, as the feature extractor Φ across all experiments. The feature extractor was followed by fully connected layers for the label classifier G and domain discriminator D . The label classifier was implemented as a simple linear layer (i.e., output layer with Softmax function) without hidden layers. For domain discriminators, we used multi-layer perceptrons with ReLU activations in their hidden layers. The DANN discriminator had one hidden layer with 256 nodes, while the CDAN discriminator employed two hidden layers with 512 and 256 nodes respectively. Note that DALN does not require a separate domain discriminator, as it reuses the task-specific classifier to compute the NWD loss. To match ResNet-18's input requirements, all images were resized to 224×224 pixels, and we used a batch size of 64 for training.

The optimization process utilized the AdamW optimizer (Loshchilov and Hutter, 2017) with an initial learning rate of 0.001. Additionally, we used the CosineAnnealingLR scheduler (Loshchilov and Hutter, 2016) to dynamically adjust the learning rate throughout the training process. We implemented different training approaches for adversarial and distribution-based UDA methods. For distribution-based UDA methods, which utilize solely the ResNet-18 architecture without a domain discriminator, we implemented early stopping with a patience of 30 epochs and a maximum limit of 100 epochs. For adversarial UDA methods, we employed a fixed training duration of 300 epochs to ensure sufficient adversarial learning, selecting the model with the lowest validation loss after a minimum of 250 epochs. This allowed for potential performance improvements in later stages of adversarial training. For hybrid distribution-adversarial method (DALN), we trained for 200 epochs in total and enabled model selection only after epoch 150, choosing the checkpoint with the lowest validation loss thereafter.

3.3. Experimental setting

We investigated the effectiveness of our proposed approach for plant disease diagnosis by evaluating how different domain adaptation methods (DDC, DCORAL, DANN, CDAN+E, DALN) combined with FBR perform when classifying real-world environment images (i.e., target domain) using models trained on laboratory images (i.e., source domain). Our experiments encompassed the following configurations:

- **Baseline:** ResNet-18 without both FBR and UDA. Trained and validated on lab-environment images.

Table 2

Performance comparison of FBR and UDA methods on real-field images. The table presents classification accuracy (%; mean $\pm$ standard deviation) over 5 iterations for both emulated and genuine settings. In the emulated setting, applying background augmentation (FBR) alone led to a significant improvement in performance across all datasets. Regarding UDA methods, both distribution-based and adversarial approaches achieved comparable performance improvements, with DDC consistently achieving the highest accuracy among all methods. In the genuine setting, while background augmentation (FBR) also contributed significantly to performance improvements, applying UDA methods, particularly adversarial approaches like CDAN+E, resulted in a substantial and meaningful enhancement in accuracy.

Exp. cases	Emulated			Genuine
	Tomato	Chili Pepper	Grape	Apple
Baseline	89.3 $\pm$ 4.18	74.0 $\pm$ 5.90	57.4 $\pm$ 6.99	47.1 $\pm$ 6.54
w/ FBR	92.8 $\pm$ 13.55	90.5 $\pm$ 3.56	81.0 $\pm$ 10.84	57.0 $\pm$ 7.01
DDC w/ FBR	98.8 $\pm$ 0.23	96.6 $\pm$ 0.50	94.2 $\pm$ 3.88	57.7 $\pm$ 3.63
DCORAL w/ FBR	98.5 $\pm$ 0.96	95.8 $\pm$ 0.98	88.1 $\pm$ 7.27	61.0 $\pm$ 3.68
DANN w/ FBR	98.3 $\pm$ 0.62	94.5 $\pm$ 1.10	90.3 $\pm$ 7.57	72.4 $\pm$ 7.50
CDAN+E w/ FBR	98.4 $\pm$ 0.45	94.0 $\pm$ 1.86	91.0 $\pm$ 3.53	91.1 $\pm$ 4.22
DALN w/ FBR	97.2 $\pm$ 0.59	91.4 $\pm$ 1.95	89.4 $\pm$ 3.53	59.3 $\pm$ 1.77
Real-to-Real	98.8 $\pm$ 1.28	97.9 $\pm$ 0.91	96.1 $\pm$ 1.44	97.7 $\pm$ 0.77

- **w/ FBR:** FBR applied during training.
- **DDC w/ FBR:** DDC with FBR during training.
- **DCORAL w/ FBR:** DCORAL with FBR during training.
- **DANN w/ FBR:** DANN with FBR during training.
- **CDAN+E w/ FBR:** CDAN+E with FBR during training.
- **DALN w/ FBR:** DALN with FBR during training.
- **Real-to-Real:** ResNet-18 trained and tested on real-field images.

Each model performance was evaluated using accuracy, and an ablation study on the FBR was also conducted to assess its individual impact on the overall performance of our proposed methods. We also established a performance upper bound by training and evaluating ResNet-18 directly on real-field images (Real-to-Real), representing ideal conditions without domain shifts. To ensure consistency and thorough evaluation, experiments were repeated 5 times, allowing for comparison of performance distributions across various real-world conditions.

Additionally, we employed two visualization techniques for qualitative analysis of the proposed methods. First, we used t-SNE (Van der Maaten and Hinton, 2008) to compare feature distributions across domains and classes by visualizing learned representations, allowing us to evaluate their characteristics in both domain adaptation and disease diagnosis. Second, we applied Grad-CAM (Selvaraju et al., 2020) to identify the key regions of input images that most influenced model predictions. These quantitative and qualitative evaluation methods offer a comprehensive assessment of our proposed approaches, providing insights into both the models' performance and behavior.

4. Results and discussion

4.1. Evaluation of domain adaptation methods

The classification performance of different models on real-field images across both emulated and genuine settings is presented in Table 2. The baseline model, trained solely on laboratory images, demonstrated limited generalization to real-field conditions, achieving accuracies of 47.1–89.3% across different crops. This performance gap was particularly pronounced in the genuine setting with apple leaf images (47.1%), while even in the emulated setting, accuracy variations were observed (57.4–89.3% across grape, chili pepper, and tomato). The application of our background recombination method (w/ FBR), which incorporates diverse backgrounds from real-field environments, yielded substantial improvements, increasing accuracy by 9.9–23.6% across all crops. Most

notably, in disease classification of grape leaf, accuracy improved from 57.4% to 81.0%.

However, despite these improvements, the performance still fell short of the Real-to-Real benchmark (96.1–98.8% accuracy), possibly due to artifacts and unnatural boundaries introduced during the background recomposition process. This gap was marked apparent in the genuine setting, where the FBR method alone achieved only 57.0% accuracy on apple leaf diseases, compared to 97.7% in the Real-to-Real scenario, highlighting the persistent challenges in bridging the domain gap between laboratory and real-field conditions. Building upon these improvements from FBR, we further examined the effectiveness of incorporating various UDA methods with FBR.

Experimental results demonstrate that this integrated approach consistently outperformed the use of FBR alone across both emulated and genuine settings. In the emulated setting characterized by controlled domain gaps, while DDC w/ FBR achieved marginally superior results (94.2–98.8% accuracy), the performance variations among different UDA methods were relatively small. In contrast, for the genuine setting with apple leaf diseases, distribution-based UDA methods showed limited effectiveness, with DDC w/ FBR achieving only 57.7% accuracy - a marginal improvement over the FBR method alone (57.0%). Notably, adversarial UDA approaches exhibited superior adaptation capabilities, with CDAN+E w/ FBR achieving the highest accuracy of 91.1%, substantially outperforming distribution-based methods (57.7–61.0% accuracy). This significant performance disparity can be attributed to the inherent differences in leaf characteristics between source and target domains, influenced by varying growth conditions, illumination, and texture variations, which present more complex domain adaptation challenges.

These results reveal that the effectiveness of domain adaptation methods is highly dependent on the complexity of the domain gap. While distribution-based UDA methods like DDC proved adequate for emulated settings with controlled domain shifts, adversarial UDA approaches were essential for addressing the more complex feature distributions encountered in genuine field conditions. To further investigate these findings, we conducted qualitative analyzes using t-SNE visualization to examine feature distributions and Grad-CAM to identify regions that influence model predictions in different domain adaptation settings.

4.2. Feature distribution visualization

To analyze the distribution of learned features across different models, we visualized their learned representations using t-SNE. Fig. 5 presents these visualizations to compare how different methods align features across domains and separate disease classes.

In the emulated setting with tomato leaf images (Fig. 5(a)), where domain shifts arise primarily from background differences, all methods applied with FBR demonstrated effective domain adaptation. They consistently achieved strong alignment between source and target domains (upper row) and clear separation among disease classes in the target domain (lower row). This indicates that the domain gap here mainly arises from background variations, which all UDA methods applied with FBR can effectively address.

The genuine setting using the PVD-PPD dataset (Fig. 5(b)) presented additional challenges in domain adaptation. In the domain alignment visualization (upper row), the baseline (without both FBR and UDA), distribution-based methods (DDC, DCORAL), and DALN showed considerable feature overlap in the target domain, indicating inadequate domain adaptation. This limitation is further reflected in the class separation visualization (lower row), where these models exhibited overlapping class clusters in their target domain representations. DANN w/ FBR showed moderate improvement in domain alignment, while CDAN+E w/ FBR demonstrated the most effective adaptation, achieving substantially improved domain alignment (upper row) and more distinct class boundaries in disease classification (lower row).

Table 3

Ablation study of the proposed method showing classification accuracy (%; mean $\pm$ standard deviation) over 5 iterations across different settings and crops. The study compares four configurations: baseline (without both FBR and UDA), individual application of FBR or UDA, and their combination. For UDA, DDC and CDAN+E were used in emulated and genuine settings, respectively. While applying either FBR or UDA individually resulted in noticeable improvements, the combination of both consistently achieved the highest accuracy, highlighting their complementary effectiveness in all settings.

Ablation	Emulated			Genuine
	Tomato	Chili Pepper	Grape	Apple
Baseline	89.3 $\pm$ 4.18	74.0 $\pm$ 5.90	57.4 $\pm$ 6.99	47.1 $\pm$ 6.54
w/ only FBR	92.8 $\pm$ 13.55	90.5 $\pm$ 3.56	81.0 $\pm$ 10.84	57.0 $\pm$ 7.01
w/ only UDA	95.5 $\pm$ 2.60	92.6 $\pm$ 1.80	84.5 $\pm$ 7.79	88.4 $\pm$ 6.57
w/ FBR & UDA	98.8 $\pm$ 0.23	96.6 $\pm$ 0.50	94.2 $\pm$ 3.88	91.1 $\pm$ 4.22

Although DDC showed strong quantitative performance in the emulated setting, its feature visualizations in the genuine setting indicate limitations in both domain alignment and class separation. This observation suggests that adversarial approaches like CDAN+E are more effective in handling domain shifts in real-world agricultural environments, where variations include changes in leaf characteristics, lighting conditions, and imaging environments.

4.3. Class activation map analysis

We also investigated how different models identify discriminative regions for disease classification by analyzing their prediction-relevant areas using Grad-CAM visualization. This analysis, focusing on the genuine setting with apple leaf images, examines whether models identify disease-relevant regions or are influenced by background elements that differ between laboratory and field conditions.

The visualizations in Fig. 6 reveal distinct patterns across different approaches. Distribution-based methods (DDC, DCORAL) and DALN tend to align domains through statistical optimization of feature distributions. However, this statistical alignment approach does not fully eliminate the influence of domain-specific elements, as evidenced by their attention to both disease-relevant areas and background regions. DALN, while incorporating some adversarial elements, relies on the task classifier itself to provide the adversarial signal rather than employing a dedicated domain discriminator, resulting in behavior closer to distribution-based approaches with limited ability to remove domain-specific influence.

In contrast, adversarial UDA methods (DANN, CDAN+E) demonstrate superior localization of disease-relevant leaf regions. These methods train feature extractors against dedicated domain discriminators, leading to the extraction of features that are both domain-invariant and disease-relevant. This adversarial mechanism effectively guides the models to focus on intrinsic disease characteristics — such as lesions and symptomatic areas — while showing minimal response to background regions that vary across domains. Among the adversarial approaches, CDAN+E achieves the most precise localization, consistently highlighting internal leaf areas containing disease symptoms with minimal background activation. This improved discriminative region identification aligns with CDAN+E w/ FBR's superior classification performance in real-field conditions, demonstrating that the adversarial approach more effectively isolates disease-specific features from domain-varying elements.

4.4. Ablation study

To systematically assess the contribution of each component in our proposed method, we conducted an ablation study in both emulated and genuine settings. We examined four different configurations: baseline (without both FBR and UDA), w/ only FBR, w/ only UDA, and

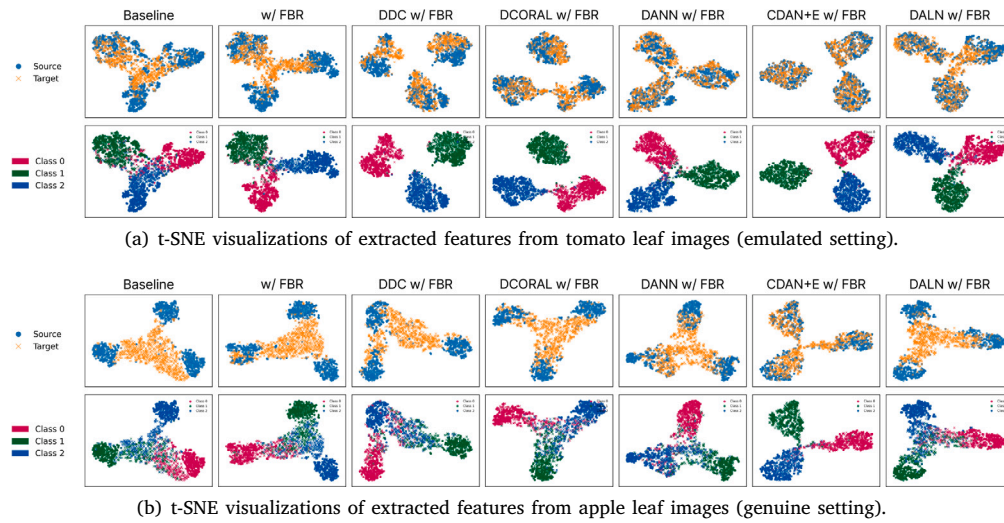


Fig. 5. t-SNE visualizations of features extracted from different models. For each subfigure, the upper rows show the distribution of source and target domain features to illustrate domain alignment, while the lower rows display the class-wise feature distribution in the target domain. (a) Feature distributions from tomato leaf data (emulated setting), where domain shift mainly stems from background variations. All UDA methods, applied in combination with FBR, generated well-aligned representations across domains with clear class-wise clustering. (b) Feature distributions from apple leaf data (genuine setting), representing more complex domain variations. Only CDAN+E with FBR achieves well-aligned source and target representations with clear class-wise clustering.

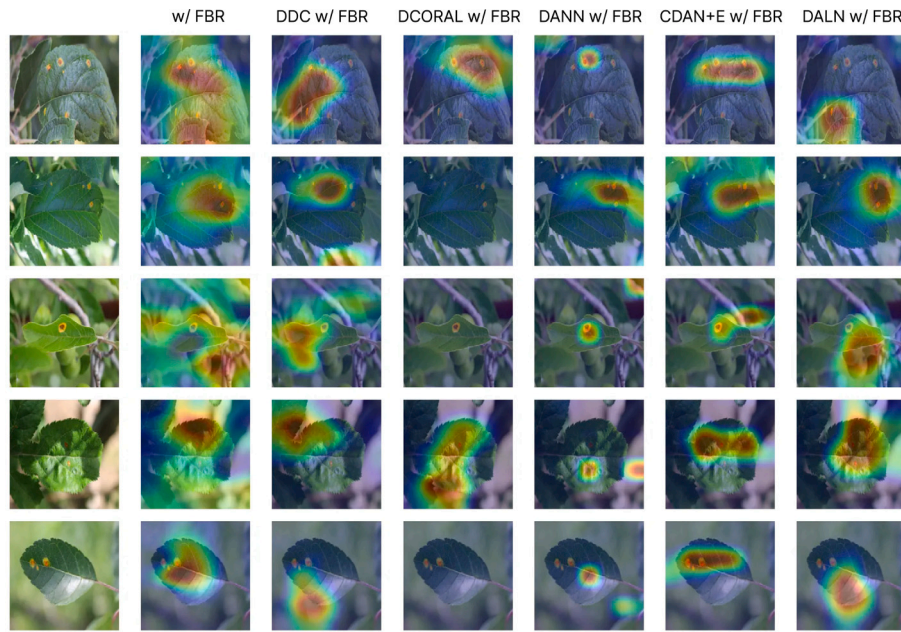


Fig. 6. Grad-CAM visualization results highlighting image regions that influence disease classifications. The heatmaps indicate regions influencing class predictions, where red indicates stronger contribution. The left-most image shows the original input for reference. Compared to the baseline model with only FBR, distribution-based approaches (DDC, DCORAL) show improved focus on lesion areas but still reflect background sensitivity. DALN provides intermediate improvements, though less sharply localized than adversarial methods. Among all methods, adversarial approaches (DANN, CDAN+E) demonstrate superior localization on disease-relevant leaf regions compared to distribution-based methods. CDAN+E shows the most precise focus on disease symptoms, while DANN occasionally highlights some irrelevant regions alongside the disease areas. All examples are randomly sampled apple leaf images from the PPD representing real-field conditions.

w/ FBR & UDA (the combination of both FBR and UDA). For the UDA configuration, we selected the most effective domain adaptation method for each setting—DDC for emulated settings and CDAN+E for genuine settings.

The results demonstrate the progressive improvement achieved by each component (Table 3). In the emulated setting, FBR alone improved accuracy by 3.5–23.6% across different crops compared to the baseline, while UDA alone showed improvements of 6.2–27.1%. The genuine

setting exhibited a similar trend, with FBR and UDA individually improving accuracy by 9.9% and 41.3%, respectively, for apple disease classification, where UDA showed notably larger improvement.

The combination of FBR and UDA consistently outperformed individual applications across all settings, demonstrating the highest accuracy improvements over the baseline in both emulated and genuine settings. These results validate that our two-step approach provides complementary benefits: FBR reduces the initial domain gap through

Table 4

Classification accuracy (%; mean $\pm$ standard deviation) for validation and test performance across different settings. The genuine setting includes a practical application case (P-Chili Pepper) for distinguishing BBWV2 viral infection from abiotic stress (flood) and healthy conditions using laboratory-generated data. Test performance for the P-Chili Pepper could not be reported due to the absence of labeled field data. Notably, the baseline model shows significant performance degradation from validation to test across all crops, indicating poor generalization. In contrast, CDAN+E w/ FBR demonstrates consistent performance between validation and test sets across all evaluated crops, suggesting strong generalization capability. The high validation accuracy (89.4%) achieved in the P-Chili Pepper indicates promising potential for effective field deployment.

Method	Split	Emulated			Genuine	
		Tomato	Chili Pepper	Grape	Apple	P-Chili Pepper
Baseline	Val	97.3 $\pm$ 10.14	95.3 $\pm$ 0.66	89.3 $\pm$ 3.56	99.9 $\pm$ 0.00	95.4 $\pm$ 1.04
	Test	89.3 $\pm$ 4.18	74.0 $\pm$ 5.90	57.4 $\pm$ 6.99	47.1 $\pm$ 6.54	N/A
w/ FBR	Val	90.8 $\pm$ 15.6	95.4 $\pm$ 12.8	89.6 $\pm$ 2.11	95.7 $\pm$ 2.52	90.2 $\pm$ 4.96
	Test	92.8 $\pm$ 13.55	90.5 $\pm$ 3.56	81.0 $\pm$ 10.84	57.0 $\pm$ 7.01	N/A
CDAN+E w/ FBR	Val	97.6 $\pm$ 1.27	92.0 $\pm$ 0.89	81.1 $\pm$ 4.79	94.6 $\pm$ 0.92	89.4 $\pm$ 1.28
	Test	98.4 $\pm$ 0.45	94.0 $\pm$ 1.86	91.0 $\pm$ 3.53	91.1 $\pm$ 4.22	N/A

background manipulation, while UDA further aligns the feature distributions, resulting in more robust disease classification models.

4.5. Practical application case

To demonstrate the practical relevance of our domain adaptation approach, we applied our FBR-UDA framework to a challenging agricultural scenario where distinguishing between viral infections and abiotic stress symptoms is critical for crop management. Plant viral diseases, such as Broad bean wilt virus 2 (BBWV2), require early detection to prevent widespread crop damage, yet differentiating viral symptoms from those caused by abiotic stresses like flooding can be challenging due to their visual similarity (Kaur and Gautam, 2021; Pantazi et al., 2017).

Using controlled laboratory conditions, we cultivated chili pepper plants under three distinct conditions: healthy, flood-stressed, and BBWV2-infected. The viral inoculation was performed through sap transmission using tobacco plants as intermediate hosts. This approach generated a total of 4102 images, comprising 1325 healthy, 1642 flood-stressed, and 1135 BBWV2-inoculated samples. These images were randomly partitioned into training (75%) and validation (25%) sets, constituting the dataset hereafter referred to as P-Chili Pepper. All images have definitive ground truth labels and were collected under biosafe conditions, preventing potential viral spread to other crops. Such controlled data generation addresses the practical challenges of field collection, where viral diagnosis requires specialized expertise and reliable symptom identification can be difficult.

Our FBR-UDA framework was applied to this three-class classification problem, where laboratory-collected images from P-Chili Pepper were adapted for field deployment using unlabeled real-field images from the AI-Hub dataset as the target domain. We evaluated the baseline model and two representative approaches: w/ FBR, which showed substantial improvements in our main experiments, and CDAN+E w/ FBR, which demonstrated the most promising generalization performance in the genuine setting.

The performance comparison is summarized in Table 4, which includes both validation and test accuracies to illustrate the generalization capability. Across all crops, the baseline model typically achieved high validation performance because the validation sets were collected under laboratory conditions with no background variation, requiring no generalization beyond the source domain. However, this high validation performance does not guarantee field deployment success, as evidenced by the significant performance degradation observed in the real-field test sets.

The w/ FBR demonstrates smaller validation-test gaps compared to the baseline. Notably, CDAN+E w/ FBR shows remarkably consistent performance between validation and test sets across all evaluated crops, indicating strong generalization capability. For P-Chili Pepper,

this consistency suggests that its validation performance (89.4%) may better reflect potential field deployment performance compared to the baseline's higher but less generalizable validation score.

While we could not evaluate test performance on actual field data for P-Chili Pepper due to the absence of verified field samples, this practical constraint illustrates a common challenge in agricultural AI: obtaining accurately labeled disease data from real farming environments is often difficult. Nevertheless, the consistent validation-test performance of CDAN+E w/ FBR observed across other crops suggests that our framework provides a potential pathway for developing diagnostic capabilities when field data collection faces significant barriers.

5. Conclusion

The scarcity of labeled real agricultural environment data limits the practical application of plant disease diagnosis models, as laboratory datasets often fail to reflect real-field conditions. This disparity hinders the deployment of AI models in real farming sites. To address this challenge, we present an effective approach for adapting plant disease classification models from laboratory to real-field environments through a novel two-step domain adaptation strategy. Our method integrates FBR as an initial domain alignment step, followed by UDA techniques, addressing the challenges of limited labeled data in agricultural settings.

The experimental results demonstrate that our approach significantly improves model performance across both emulated and genuine field conditions. In the emulated setting, where domain shifts primarily arise from background variations, both distribution-based and adversarial UDA methods achieved comparable high performance when enhanced by FBR. However, in the genuine setting with more complex domain disparities, adversarial approaches, particularly CDAN+E with FBR, demonstrated superior adaptation capabilities, achieving 91.1% accuracy compared to 57.7% with distribution-based methods. Feature visualization and activation map analyses revealed that adversarial UDA methods more effectively isolate disease-relevant features from domain-specific variations, particularly in real-field conditions. The ablation study further confirmed the complementary benefits of our two-step approach, where FBR reduces initial domain gaps through background manipulation, while UDA further aligns feature distributions for robust disease classification.

Our findings suggest that this approach can effectively bridge the gap between laboratory-trained models and their practical deployment in agricultural environments, particularly when dealing with complex domain shifts involving leaf characteristics, lighting conditions, and imaging environments. The ability of this method to utilize readily available unlabeled field images makes it particularly valuable for real-world agricultural applications, where obtaining labeled data from diverse field conditions remains challenging.

Limitations and future directions. While our proposed FBR-UDA framework effectively addresses key aspects of the Lab-to-Field gap, several limitations should be acknowledged. Our analysis primarily focused on background variations and basic environmental factors, but lacked systematic examination of the multidimensional nature of domain discrepancies. The Lab-to-Field gap encompasses additional dimensions that were not comprehensively analyzed, including dynamic lighting changes, structural occlusions from overlapping leaves, heterogeneous background clutter from farming equipment and infrastructure, and physiological variations in leaf states due to different growth conditions, nutritional status, and maturity stages. Future work should develop methodologies that address these more complex domain variations through comprehensive data collection and analysis across diverse environmental conditions and agricultural settings. Additionally, extending our approach to advanced imaging modalities such as near-infrared (NIR) or hyperspectral imaging could enhance the detection of subtle symptom differences that are challenging to distinguish in RGB images alone, particularly for early-stage infections or conditions with similar visual manifestations (Nguyen et al., 2021; Wang et al., 2024; Zhu et al., 2024).

CRedit authorship contribution statement

Woosang Jeon: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Taehyeong Kim:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Sanghyeok Choi:** Writing – review & editing, Validation. **Kyuseok Yang:** Writing – review & editing, Validation. **Seong-Yeop Kim:** Writing – review & editing, Validation, Data curation. **Mungyeong Song:** Writing – review & editing, Validation, Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-4o, ChatGPT-5 and Claude 3.7 Sonnet in order to assist with English translation and refinement. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the New Faculty Start-up Fund and the Creative-Pioneering Researchers Program through Seoul National University. Additionally, this research was supported by the following grants: the National Research Foundation of Korea, South Korea (RS-2023-00302123) funded by the Korean government, as part of the European Commission's Horizon Europe framework programme (Grant Agreement No. 101135576, INTEND); the Korea Institute of Planning and Evaluation for Technology in Food, Agriculture, and Forestry (RS-2024-00398300) through the Agriculture and Food Convergence Technologies Program for Research Manpower Development; and the R&D Program for Forest Science Technology (RS-2024-00402509) of the Korea Forest Service (Korea Forestry Promotion Institute).

The authors would like to thank Professor Jang-Kyun Seo and the Plant Molecular Immunology Lab at Seoul National University for providing the infectious clone of BBWV2 and plant materials. This research utilized datasets from The Open AI Dataset Project (AI-Hub, S. Korea), and all data information can be accessed through AI-Hub (www.aihub.or.kr).

Data availability

All datasets used in this study are publicly available as referenced in the manuscript. The P-Chili Pepper dataset developed in this study is available at <https://doi.org/10.5281/zenodo.17901489>, and the source code and experimental protocols are available at <https://github.com/LABA-SNU/FBR-UDA> to ensure complete reproducibility.

References

- Chen, L., Chen, H., Wei, Z., Jin, X., Tan, X., Jin, Y., Chen, E., 2022. Reusing the task-specific classifier as a discriminator: Discriminator-free adversarial domain adaptation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 7181–7190.
- Elumalai, M., Fernandez, T.F., Ragab, M., 2024. Machine learning (ML) algorithms on IoT and drone data for smart farming. In: *Intelligent Robots and Drones for Precision Agriculture*. Springer, pp. 179–206.
- Fan, X., Luo, P., Mu, Y., Zhou, R., Tjahjadi, T., Ren, Y., 2022. Leaf image based plant disease identification using transfer learning and feature fusion. *Comput. Electron. Agric.* 196, 106892.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., March, M., Lempitsky, V., 2016. Domain-adversarial training of neural networks. *J. Mach. Learn. Res.* 17 (59), 1–35.
- Grinblat, G.L., Uzal, L.C., Larese, M.G., Granitto, P.M., 2016. Deep learning for plant identification using vein morphological patterns. *Comput. Electron. Agric.* 127, 418–424.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 770–778.
- Hughes, D., Salathé, M., et al., 2015. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060*.
- Ilyas, T., Lee, J., Won, O., Jeong, Y., Kim, H., 2023. Overcoming field variability: unsupervised domain adaptation for enhanced crop-weed recognition in diverse farmlands. *Front. Plant Sci.* 14, 1234616.
- Kaur, P., Gautam, V., 2021. Research patterns and trends in classification of biotic and abiotic stress in plant leaf. *Mater. Today: Proc.* 45, 4377–4382.
- Kaushik, M., Prakash, P., Ajay, R., Veni, S., et al., 2020. Tomato leaf disease detection using convolutional neural network with data augmentation. In: *2020 5th International Conference on Communication and Electronics Systems. ICCES, IEEE*. pp. 1125–1132.
- Kim, T., Lee, D.-H., Kim, W.-S., Zhang, B.-T., 2023. Domain adapted broiler density map estimation using negative-patch data augmentation. *Biosyst. Eng.* 231, 165–177. <http://dx.doi.org/10.1016/j.biosystemseng.2023.06.006>, URL: <https://www.sciencedirect.com/science/article/pii/S1537511023001241>.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., et al., 2023. Segment anything. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 4015–4026.
- Liu, J., Wang, X., 2020. Tomato diseases and pests detection based on improved yolo V3 convolutional neural network. *Front. Plant Sci.* 11, 898.
- Liu, J., Wang, X., 2021. Plant diseases and pests detection based on deep learning: a review. *Plant Methods* 17, 1–18.
- Long, M., Cao, Z., Wang, J., Jordan, M.I., 2018. Conditional adversarial domain adaptation. *Adv. Neural Inf. Process. Syst.* 31.
- Loshchilov, I., Hutter, F., 2016. Sgdr: Stochastic gradient descent with warm restarts. *arXiv preprint arXiv:1608.03983*.
- Loshchilov, I., Hutter, F., 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.
- Van der Maaten, L., Hinton, G., 2008. Visualizing data using t-SNE. *J. Mach. Learn. Res.* 9 (11).
- Mahlein, A.-K., 2016. Plant disease detection by imaging sensors—parallels and specific demands for precision agriculture and plant phenotyping. *Plant Dis.* 100 (2), 241–251.
- Marino, S., Beauserey, P., Smolarz, A., 2020. Unsupervised adversarial deep domain adaptation method for potato defects classification. *Comput. Electron. Agric.* 174, 105501. <http://dx.doi.org/10.1016/j.compag.2020.105501>, URL: <https://www.sciencedirect.com/science/article/pii/S0168169919320903>.
- Martinelli, F., Scalenghe, R., Davino, S., Panno, S., Scuderi, G., Ruisi, P., Villa, P., Stroppiana, D., Boschetti, M., Goulart, L.R., et al., 2015. Advanced methods of plant disease detection. A review. *Agron. Sustain. Dev.* 35, 1–25.
- Mohanty, S.P., Hughes, D.P., Salathé, M., 2016. Using deep learning for image-based plant disease detection. *Front. Plant Sci.* 7, 1419.
- Nguyen, C., Sagan, V., Maimaitiyiming, M., Maimaitijiang, M., Bhadra, S., Kwasniewski, M.T., 2021. Early detection of plant viral disease using hyperspectral imaging and deep learning. *Sensors* 21 (3), 742.
- Pantazi, X.E., Moshou, D., Oberti, R., West, J., Mouazen, A.M., Bochtis, D., 2017. Detection of biotic and abiotic stresses in crops by using hierarchical self organizing classifiers. *Precis. Agric.* 18 (3), 383–393.

- Ramesh, S., Hebbar, R., M., N., R., P., N., P.B., N., S., P.V., V., 2018. Plant disease detection using machine learning. In: 2018 International Conference on Design Innovations for 3Cs Compute Communicate Control. ICDI3C, pp. 41–45. <http://dx.doi.org/10.1109/ICDI3C.2018.00017>.
- Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2020. Grad-CAM: visual explanations from deep networks via gradient-based localization. *Int. J. Comput. Vis.* 128, 336–359.
- Shi, M., Li, X.-Y., Lu, H., Cao, Z.-G., 2022. Background-aware domain adaptation for plant counting. *Front. Plant Sci.* 13, 731816.
- Shoaib, M., Hussain, T., Shah, B., Ullah, I., Shah, S.M., Ali, F., Park, S.H., 2022. Deep learning-based segmentation and classification of leaf images for detection of tomato plant disease. *Front. Plant Sci.* 13, 1031748.
- Sun, B., Feng, J., Saenko, K., 2016. Return of frustratingly easy domain adaptation. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 30, 1.
- Sun, B., Saenko, K., 2016. Deep coral: Correlation alignment for deep domain adaptation. In: *Computer Vision—ECCV 2016 Workshops: Amsterdam, the Netherlands, October 8–10 and 15–16, 2016, Proceedings, Part III* 14. Springer, pp. 443–450.
- Syed-Ab-Rahman, S.F., Hesamian, M.H., Prasad, M., 2022. Citrus disease detection and classification using end-to-end anchor-based deep learning model. *Appl. Intell.* 52 (1), 927–938.
- Tzeng, E., Hoffman, J., Zhang, N., Saenko, K., Darrell, T., 2014. Deep domain confusion: Maximizing for domain invariance. *arXiv preprint arXiv:1412.3474*.
- Wang, Y., Kootstra, G., Yang, Z., Khan, H.A., 2024. UAV multispectral remote sensing for agriculture: A comparative study of radiometric correction methods under varying illumination conditions. *Biosyst. Eng.* 248, 240–254.
- Wu, X., Fan, X., Luo, P., Choudhury, S.D., Tjahjadi, T., Hu, C., 2023. From laboratory to field: Unsupervised domain adaptation for plant disease recognition in the wild. *Plant Phenomics* 5, 0038.
- Yan, K., Guo, X., Ji, Z., Zhou, X., 2021. Deep transfer learning for cross-species plant disease diagnosis adapting mixed subdomains. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 20 (4), 2555–2564.
- Yashodha, G., Shalini, D., 2021. An integrated approach for predicting and broadcasting tea leaf disease at early stage using IoT with machine learning—a review. *Mater. Today: Proc.* 37, 484–488.
- Yu, Y., Wu, X., Yu, P., Wan, Q., Dan, Y., Xiao, Y., Wang, Q., 2025. Location-guided lesions representation learning via image generation for assessing plant leaf diseases severity. *Plant Phenomics* 100058.
- Zhao, K., Wu, X., Xiao, Y., Jiang, S., Yu, P., Wang, Y., Wang, Q., 2024. PlanText: Gradually masked guidance to align image phenotypes with trait descriptions for plant disease texts. *Plant Phenomics* 6, 0272.
- Zhao, T., Zhang, G., Zhong, P., Shen, Z., 2023. DMDnet: A decoupled multi-scale discriminant model for cross-domain fish detection. *Biosyst. Eng.* 234, 32–45. <http://dx.doi.org/10.1016/j.biosystemseng.2023.08.012>, URL: <https://www.sciencedirect.com/science/article/pii/S1537511023001733>.
- Zhu, H., Cen, H., Zhang, C., He, Y., 2016. Early detection and classification of tobacco leaves inoculated with tobacco mosaic virus based on hyperspectral imaging technique. In: 2016 ASABE Annual International Meeting. American Society of Agricultural and Biological Engineers, p. 1.
- Zhu, H., Chu, B., Zhang, C., Liu, F., Jiang, L., He, Y., 2017. Hyperspectral imaging for presymptomatic detection of tobacco disease with successive projections algorithm and machine-learning classifiers. *Sci. Rep.* 7 (1), 4125.
- Zhu, H., Lin, C., Liu, G., Wang, D., Qin, S., Li, A., Xu, J.-L., He, Y., 2024. Intelligent agriculture: Deep learning in UAV-based remote sensing imagery for crop diseases and pests detection. *Front. Plant Sci.* 15, 1435016.